Note that, however, [8, Proposition 10.10(2)] does not hold. One of the main motivations of this paper is to construct an alternative to be used in applications.

Notation 3.19. For simplicity, we often drop the constant volume form 0 from the notation. For instance, if $\boldsymbol{U}$ is an object of $\mathrm{RV}[*]$ then it may also denote the object $(\boldsymbol{U}, 0)$ of $\mu\mathrm{RV}[*]$ with the constant volume form 0.

Notation 3.20. Let $[1] \in \mathbf{K}_+\,\mathrm{RES}[1]$ be the class of the singleton $\{1\}$. The class of the singleton $\{1\}$ in $\mathbf{K}_+\,\mathrm{RES}[0]$ is the multiplicative identity of $\mathbf{K}_+\,\mathrm{RES}[*]$ and hence is simply denoted by 1. We have $[\mathrm{RV}^{\circ\circ}_{\infty}] = [\mathrm{RV}^{\circ\circ}] + 1$ in $\mathbf{K}_+\,\mathrm{RV}[\leq 1]$.

Let I_{sp} be the (nonhomogeneous) semiring congruence relation on $\mathbf{K}_+\,\mathrm{RV}[*]$ generated by the pair $([1], [\mathrm{RV}^{\circ\circ}_{\infty}])$. Let

$$\boldsymbol{P} = [1] - [\mathrm{RV}^{\circ\circ}] \in \mathbf{K}\,\mathrm{RV}[1].$$

The corresponding principal ideal of $\mathbf{K}\,\mathrm{RV}[*]$ is thus generated by the element $\boldsymbol{P} - 1$.

Similarly, let $\mu\mathrm{I}_{\mathrm{sp}}$ be the semiring congruence relation on $\mathbf{K}_+\,\mu\mathrm{RV}[*]$ generated by the pair $([1], [\mathrm{RV}^{\circ\circ}])$, which is homogeneous, and the corresponding principal ideal of $\mathbf{K}\,\mu\mathrm{RV}[*]$ is generated by the element $\boldsymbol{P}$.

Resolved referee R051 [Typo] – Spelling of homogeneous.
Original source line 1358. In this notation, replace "nonhomogenous" and "homogenous" by "nonhomogeneous" and "homogeneous," respectively.

Author resolution. The author approves correcting the spellings to nonhomogeneous and homogeneous. The congruences and their generators are unchanged.

Notation 3.21. For each $\boldsymbol{U} = (U, f) \in \mathrm{RV}[k]$, let U_f be the set $\bigcup\{f(u)^\sharp \times u \mid u \in U\}$. Let $\mathbb{L}_{\leq k} : \mathrm{RV}[\leq k] \longrightarrow \mathrm{VF}[k]$ be the map given by $\boldsymbol{U} \longmapsto U_f$. Set $\mathbb{L} = \bigcup_k \mathbb{L}_{\leq k}$.

Let $\mu\mathbb{L}_k : \mu\mathrm{RV}[k] \longrightarrow \mu\mathrm{VF}[k]$ be the map given by $(\boldsymbol{U}, \omega) \longmapsto (\mathbb{L}\boldsymbol{U}, \mathbb{L}\omega)$, where $\mathbb{L}\omega$ is the obvious function on $\mathbb{L}\boldsymbol{U}$ induced by ω. Set $\mu\mathbb{L} = \bigoplus_k \mu\mathbb{L}_k$.

Suppose that $\mathbb{S}$ is (VF, Γ)-generated. The map $\mathbb{L}$ induces a surjective semiring homomorphism $\mathbf{K}_+\,\mathrm{RV}[*] \longrightarrow \mathbf{K}_+\,\mathrm{VF}_*$ and the map $\mu\mathbb{L}$ induces a surjective graded semiring homomorphism $\mathbf{K}_+\,\mu\mathrm{RV}[*] \longrightarrow \mathbf{K}_+\,\mu\mathrm{VF}[*]$, see [8, § 4, § 6] or [17, Corollaries 7.7, 10.6]; we use the same notation for these homomorphisms as well as their groupifications. We have

Resolved referee R053 [Error; revised] – Scope of the parameter hypothesis.
Original source line 1366. State here the (VF, Γ)-generation hypothesis used in Theorem 3.22, before asserting that lifting induces these homomorphisms. Without a parameter restriction, even well-definedness can fail. For example, let the parameters be generated by a trivially valued algebraically closed field C and a named $\tau \in \mathrm{RV}$ of positive value. The singleton objects $\{\tau\}$ and $\{1\}$ are isomorphic in $\mathrm{RV}[1]$. But $\tau^\sharp$ has no definable VF point: Lemma 2.13 places every such point in $\mathrm{acl}(C) \cap \mathrm{VF} = C$. In contrast, $1 + \mathcal{M}$ contains the definable point 1, so their lifts cannot be definably bijective. Moving the stated hypothesis forward removes this obstruction.

Author resolution. The author approves stating the (VF, Γ)-generation hypothesis before both homomorphism assertions. The lift on individual objects needs no such hypothesis. Under the standing nontrivial-ground-value-group assumption, take instead a small algebraically closed nontrivially valued subfield F and name $\tau \in \mathrm{RV}$ with $\mathrm{vrv}(\tau) \notin \Gamma(F)$. Lemma 2.13 puts every definable VF point over the generated structure in F. Thus $\tau^\sharp$ has no definable point, whereas $1^\sharp$ contains 1, although the singleton objects are isomorphic in $\mathrm{RV}[1]$.

$$\mathbb{L}([1]) = [1 + \mathcal{M}] = [\mathcal{M}] \quad \text{and} \quad \mathbb{L}([\mathrm{RV}^{\circ\circ}]) = [\mathcal{M} \smallsetminus 0],$$